



Electronic Catalogs: a Technology Overview and Survey Results

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Abstract

Electronic catalogs are the front-end to the rapidly evolving Internet-based global marketplace, and the gateway to a company through which customers can obtain product information, order goods and services, make payments, access customer support, and provide feedback around the clock and from anywhere in the world. This paper presents an overview of electronic catalogs technology and the results from a three-phase survey of over 100 companies. The study results indicate that most existing catalogs provide only rudimentary functionalities, such as embedded graphics and simple browsing. Companies are successful in learning about the technology and in enhancing corporate image, but are not as successful in selling products. The two most important measures of catalog success are the number of visits per month and the percentage of visits resulting in sales, but the most widely used measure is the number of monthly visits. Two factors are found to have noticeable impact on the number of monthly visits: the level of overall investment in the catalog and the type of catalogs, i.e., standalone, mall, or embedded.

1 Introduction

The Internet is rapidly transforming the way companies conduct their business. It is not only opening vast new opportunities but at the same time drastically changing existing business practices. With the advent of the World Wide Web (WWW), the Internet has become a platform that an increasing number of companies turn to for product and process innovations. The popular "Yahoo" index, for instance, contains pointers to over 20,000 commercial sites, many of which are virtual storefronts and provide products and services that can be ordered directly online, and paid for via credit cards or other online payment methods. While the Internet has not yet become a mass market from which companies can actually derive sizeable revenues, the chance that this will become a reality is quite high. Hence, the current trend in distributing product and company information to customers and transacting business electronically is likely to

continue and perhaps accelerate.

Electronic commerce refers to paperless business transactions which use technologies such as electronic data interchange (EDI), electronic mail, electronic funds transfer (EFT), electronic bulletin boards, and so on. Conceptually, it is useful to view electronic commerce as being comprised of two parts: "front-end" and "back-end." A front-end application interacts directly with a human user, such as a consumer or an engineer. Many Web-based applications, such as electronic catalogs and online malls, fall into this category. A back-end application, on the other hand, accepts data from one computer application, processes it, and pipes the result into another application with no human intervention. Traditional EDI falls into this category. The back-end and front-end components are both required to provide end-to-end, transparent business-to-business and business-to-consumer transactions.

While electronic commerce has been around in a limited scope for a number of years, large-scale, open electronic commerce is still a recent phenomenon. One primary driving force behind this open electronic commerce movement is the Internet – the world's largest and fastest growing public network. With about 30 million users from over 100 countries and with an annual growth rate of over 100% each year, the Internet provides the only viable network infrastructure that can serve as the backbone of a global electronic marketplace, linking together businesses and consumers, large and small companies, and foreign and domestic enterprises. Furthermore, the Internet's open architecture, standards, and culture encourage collaboration, experimentation, and entrepreneurship. The explosive growth of the Internet as a marketplace during the last few years demonstrates the tremendous benefits of open electronic commerce. For example, Internet software providers, such as Netscape Communications, Sun Microsystems, and Silicon Graphics, rely heavily on the Internet itself to market and distribute their products, solicit user feedback, provide customer support, and facilitate corporate communications. The result is a significantly shortened development cycle and reduced cost largely due to online delivery of information. Small businesses such as single-person Internet consulting companies benefit especially from this environment because of the low-cost, interactive, and global nature of the Internet.

This paper focuses on a front-end electronic commerce application, namely, *electronic catalogs*. First, it defines what an electronic catalog is, and provides an overview of functional features of electronic catalogs and the current state of technology. Next, it describes our survey study on the current practice of electronic catalogs, including the

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conceptual framework, methodology and results. Finally, the paper identifies several future trends and promising research directions in the electronic catalogs technology.

2 Overview of Electronic Catalog Technology

2.1 What is an Electronic Catalog?

In this paper, the term “electronic catalog” is defined broadly to include any World Wide Web (WWW) page that contains information about the products and/or services a commercial entity offers. An electronic catalog does not have to contain detailed pricing information or provide online ordering and payment capabilities. However, it should provide pointers on how to obtain such information, how to order products/service, and how to pay for the products/services ordered. A Web site can contain one or more catalogs. An “electronic mall,” for example, is a collection of catalogs from different companies who each rent “virtual store space” from a common service provider. The terms “electronic storefronts” and “Web catalogs” are used synonymously with “electronic catalogs.”

It is important to note that an electronic catalog is not merely an electronic replica of a printed catalog. Rather, it is an entirely new medium that can be more appropriately viewed as the virtual “gateway” or “front-door” into a company. Electronic catalogs possess several important characteristics, including *interactivity*, *dynamic updating*, *hypertextuality*, and *global presence*:

- **Interactivity:** Printed catalogs represent a one-way flow of information from the company to its potential customers. Electronic catalogs, in contrast, provide a two-way communication channel between the company and its customers. For example, if a customer has a question or comment about a product while viewing its catalog, he or she can simply click on the e-mail button and enter his or her comments. The feedback is instantaneous, contextual, and almost effortless. This ability to carry out “dialogues” with potential customers enables the company to develop closer relationships with its customers and be more responsive to their needs.
- **Dynamic updating:** Electronic catalogs are stored in a centralized server and accessed by client browsers distributed throughout the world. The catalog content can be easily and constantly updated on the server, and the update is instantaneously visible to all clients. This dynamic nature of electronic catalogs enables companies to rapidly respond to changing market conditions through repackaging, re-pricing, and so on.
- **Hypertextuality:** The Internet represents a rich repository of information resources, ranging from anonymous FTP files and gopher documents to newsgroups. The World Wide Web, on which electronic catalogs are built, provides an elegant way of linking together related resources. For example, a wine catalog might contain not only such essential information as taste and price but also pointers to information such as the winery that makes a particular wine, recipes that go with the wine, and third-party wine reviews. These hyperlinked resources form value-added content which user may choose to navigate when needed.
- **Global presence:** The Internet is a global network of networks. Hence, any electronic catalog is a “shopping

stop” in a global marketplace that transcends the traditional limitations of time and space. Since the cost for establishing such a global presence is minimal compared to that required to print a traditional paper catalog, electronic catalogs offer small and medium-sized companies a special advantage in competing globally.

2.2 Catalog Features

Since electronic catalogs are essentially the gateway to a company, they are used to serve multiple business objectives, including marketing, sales, distribution and customer support. As a result, electronic catalogs vary widely in their functionality. A simple catalog, for example, may contain only product descriptions and pricing information, with no searching, online ordering or payment capabilities. A full-fledged catalog, in contrast, can be fully integrated with the company’s product databases. Users can conduct sophisticated searches using product-specific criteria, place orders in real-time, and make payment directly through secured online mechanisms.

Catalog features can be divided into five groups: *content*, *presentation*, *back-end processing*, *computational services*, and *usage tracking*:

- **Content:** what is the scope of the catalog: product, service, or both? How detailed is the descriptive and pricing information? How current is the catalog content? Does the catalog contain links to related company information, such as news and personal homepages? Does it contain links to related third-party resources, such as product reviews and USENET discussions?
- **Presentation:** What multimedia capabilities does the catalog use: embedded graphics, audio, video? Does it use any interface metaphors (e.g., “shopping cart”)? Does it use any 3-D techniques?
- **Back-End processing:** Is the catalog integrated with the company’s product database? Does the catalog have built-in interfaces with legacy systems such as ordering and accounts receivable?
- **Computational services:** What search capabilities does the catalog support: browsing only, Boolean, WAIS, fuzzy, product-specific? Does it support cross-searching of multiple product databases? Does it provide online ordering capabilities? What secured payment mechanisms does the catalog support? Does the catalog allow “test runs” and/or direct downloading of a product? What types of feedback mechanisms does the catalog provide: e-mail, chatting, voice mail, live video?
- **Usage tracking:** Does the catalog keep track of the people who visit the site, that is, does the catalog require registration? What about the number of visits and the amount of time spent during each visit? Can the catalog automatically customize its presentation based upon its current knowledge of the user?

2.3 Three Types of Catalogs

Electronic catalogs can be divided into three main types based upon the Web context in which they are located: *standalone*, *electronic mall*, and *embedded*. A standalone catalog is an independent Web site that is established with the explicit purpose of selling selected products or services. A

good example of such a catalog is Virtual Vineyards, shown in Figure 1. The Virtual Vineyards catalog is analogous to an electronic specialty store except for the absence of a human clerk. For each listed wine, it contains a detailed description and pricing information, including volume discounts and other special offers. The catalog contains a rich set of links to related information, such as the winery from which a given wine is made. While the catalog does not permit any advanced searching, it does provide access to the same wine from many different access points, such as wine type, price range, or winery. Perhaps the most noticeable feature of this catalog is its sophisticated online ordering and payment function, including secured credit card and CyberCash options.

An electronic mall is a collection of catalogs belonging to a number of vendors who share the same virtual space provided by a single network access provider. Like standalone catalogs, electronic malls are set up with the explicit purpose of selling products or services. Since a mall consists of many catalogs, its content tends to vary widely from catalog to catalog. In general, however, the presentation style of all catalogs in a mall is consistent. A unique feature of malls is that they have a directory that serves as the top-level entry point, as shown in Figure 2. Similar to a physical mall, an electronic mall may also provide services such as entertainment centers, information desks, and community bulletin boards.

Embedded catalogs are the catalogs that exist as part of a larger corporate Web site. One major difference between them and their standalone and mall counterparts is that these catalogs are not necessarily designed to support direct sales. Rather, they might be used to help accomplish other corporate objectives, including marketing, customer support, and communications. At the content level, embedded catalogs usually contain more detailed description about related products or services, and a rich set of links to other company information. Figure 3 shows a product catalog from the Hewlett-Packard corporate Web site.

3 The Survey Study

3.1 Objectives

In an attempt to understand the current practice of electronic catalogs, the Fisher Center for Information Technology and Management (CITM) at University of California, Berkeley, initiated a survey study entitled "Designing Electronic Catalogs for Business Value." The primary objective of this study is to gather preliminary data that can help answer a number of important questions: what types of catalogs are in the existing Web world? What functional features do they have in common? Who are the companies that are investing in electronic catalogs? What business objectives do they want to accomplish through such catalogs? What measures do they use to evaluate the success or failure of electronic catalogs?

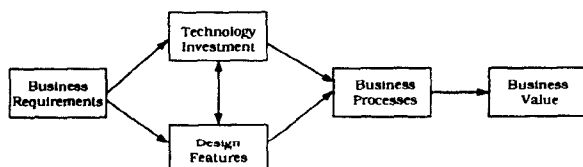


Figure 4: A Framework for Studying Electronic Catalogs.

As depicted in Figure 4, our approach to electronic catalogs is broad, encompassing both technological and business factors: business requirements, levels of investment, functional features, business processes, and business outcomes. The basic premise of this approach is that all these factors are interdependent, and thus must be studied in context. Our long-term objective is to uncover evidence that might support any relationships existing among these factors, for instance, how the level of investment affects functional features, and how functional features in turn contribute to the business outcomes. The current study, however, represents merely a bootstrapping effort. Thus, it may not provide enough data to fully answer any of these questions.

3.2 Methodology

This study consists of three parts: *functional*, *technical*, and *business*. The functional component focuses on two important questions: what electronic catalogs are already there in the Web world? What functional features do they have? To answer these questions, a total of 508 commercial Web sites were randomly selected from well-known directories, including Yahoo, the Commercial Site Index from OpenMarket, and CommerceNet membership directory. These selected sites were examined by one of the research team members using Netscape or Mosaic, and a detailed survey data form was filled out for each site.

The technical component of the study involved surveying the developers of the selected catalogs. This part focuses on the requirements of catalogs, the extent to which those requirements are translated into functional features, and costs. The technical survey was conducted via a combination of telephone interviews, e-mail, fax, and online Web forms.

The business survey was targeted at the business managers who were responsible for the investment and outcomes of the selected catalogs, such as division managers or heads of marketing department. This step focused on the impact of electronic catalogs on business processes, performance in terms of perceived values and actual outcomes, and actual costs. Like the technical survey, the business survey was also conducted via a combination of telephone interviews, e-mail, fax, and online Web forms.

This study officially began in February, 1995. Most of the functional surveys were done during March and April. The technical and business ones were carried out concurrently between May and early June.

3.3 Overview of the Results

Of the 508 sampled Web sites, 364 were identified as catalogs according to the definition given in Section 2.1. After contacting these 364 companies via telephone and e-mail, over 100 of them responded to the technical and/or business surveys. Below are highlights of findings from the survey:

- As shown in Figure 5, electronic catalogs are used by a wide range of industries, but most are in retailing, entertainment, consulting, and software.
- Electronic catalogs are not limited to information products or services. In fact, more often, physical products such as computers are listed and sold.
- Quite surprisingly, about 66% of the sampled catalogs have T-1 or faster Internet connections.

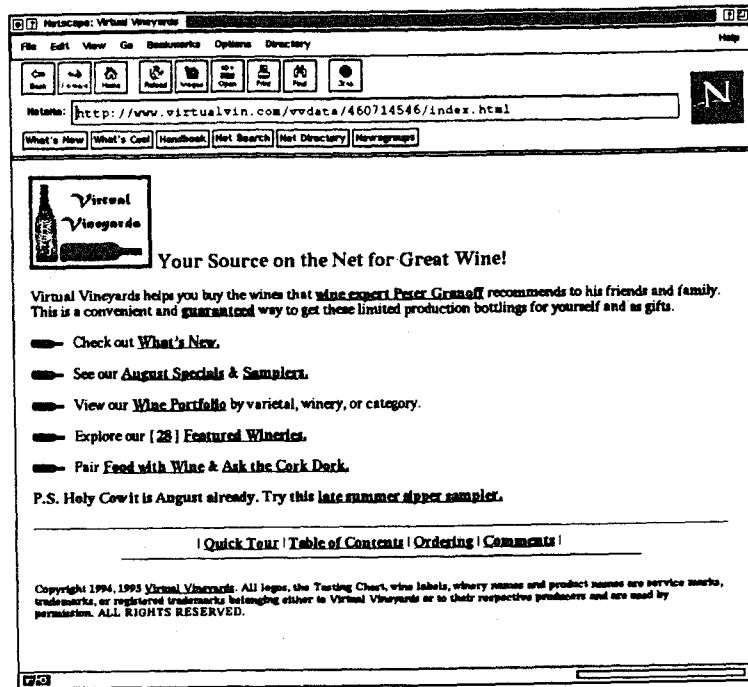


Figure 1: An Example of the Standalone Catalog: Virtual Vineyards.

- A majority of surveyed companies (75%) relied on their internal staff to develop their catalogs.

3.4 Functionality of Existing Catalogs

The survey shows that existing electronic catalogs offer quite limited functionalities. A vast majority of catalogs, for instance, provide only simple browsing capability. As shown in Figure 6, even the expectations from business managers on electronic catalogs are not high. The top three functional requirements include professional-quality page design and layout, the ability to track usage, and support for interactive feedback. Most companies do provide interactive feedback mechanisms through their Web catalogs to allow customers to directly interact with the company. Surprisingly, however, 45% of the catalogs support online ordering, with an additional 27% planning to offer this capability in the near future. Furthermore, 25% of the surveyed catalogs support online payment, with an additional 34% planning to in the near future.

At the content level, over half of the catalogs (55%) contain pricing information detailed enough to make purchasing decisions. While many companies use the Web to link their catalogs to related online company information, such as news releases, few catalogs are linked to external online resources, such as USENET newsgroups and third-party product reviews. Furthermore, few of the existing catalogs make full use of the Web multimedia capabilities except for embedded graphics.

3.5 Business Practices

The survey reveals that top management is among the main initiators of electronic catalogs (61%), and that over half of the catalogs were initiated (57%) and owned (71%) primarily

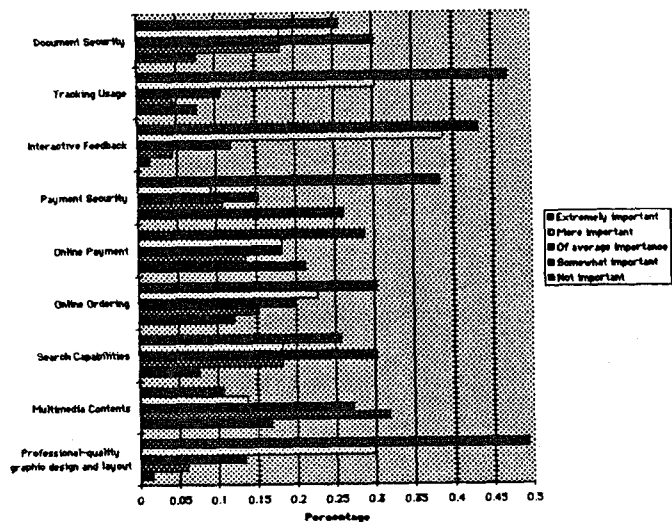


Figure 6: Functional Requirements of Catalogs from Business Managers.

by the marketing and sales departments. The average electronic catalog attracts many visitors but few purchasers. Nearly half (47%) of the catalogs reported that 1% or fewer of their visits resulted in sales. There were three companies which indicated that up to 20% of the visits to their catalogs resulted in a sale, and all three of them were software companies.

The low-cost advantage of the Internet is clearly evident in electronic catalogs. Nearly 75% of the catalogs have an initial setup cost of \$10,000 or less. About 70% of the cata-

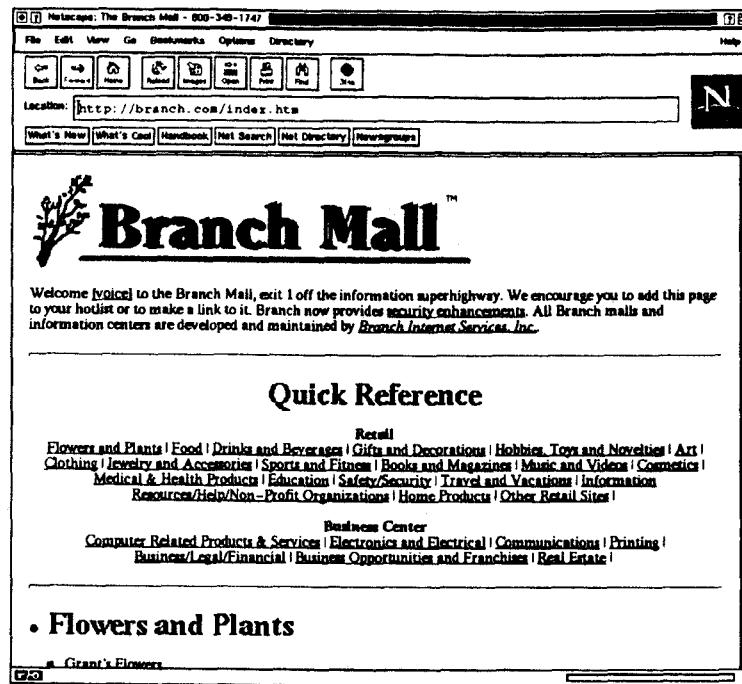


Figure 2: An Example of the Electronic Mall: Branch Mall.

logs have a monthly maintenance cost of \$1,000 or less. The two main cost components of catalogs are content creation and telecommunications links, which account for 37% and 25% of the total cost, respectively.

The survey results also indicate that the catalog owners see no substantial obstacles to fully exploiting the potential of electronic catalogs. The two biggest barriers mentioned were that the potential customers did not all have Internet access (average weight of 3.02 on a scale of 1-5, with 5 being the largest barrier) and concerns about the lack of security on the Internet (average score of 2.62 on the scale of 1 to 5). With respect to the security question, many respondents stressed that it was the customer's perception rather than an actual lack of security that was the problem. One manager pointed out that when a waiter in a restaurant takes a credit card out of sight, the credit card is no less secure than it would be over the Internet; it is merely that a restaurant is more familiar than the Internet and the customer feels more secure.

3.6 Measuring the Business Value

As mentioned earlier, electronic catalogs are a versatile application that can be used to help accomplish a wide range of business objectives, such as providing a new marketing channel, enhancing corporate or brand image, selling products or services, reducing marketing and distribution cost, improving customer support, and learning about the Internet technology. To evaluate the degree to which each of these objectives is achieved, measures must be introduced, and it is likely that each objective requires different measure(s).

The survey results indicate that the three most important business objectives of electronic catalogs are:

- Providing a new marketing channel;

- Enhancing corporate or brand image; and
- Selling products and/or services.

Of the measures identified in the survey, the number of visits per month and the percentage of visits resulting in sales are considered most important. However, it is the number of visits that is the single most frequently used measure: 81% of the catalog owners said they used this measure as a measure of success, though only about half of them (56%) were actually able to tell us how many visits they had received in the past month. A visit is defined as a single user browsing the catalog; a single user who looks at the catalog twice, for instance, would count as two visits.

Similarly, 81% of the catalog owners said they used the percentage of catalog visits resulting in a sale as a measure of success. Yet, only 43% were able to identify what percentage was. Only one company in the sample had set up a separate tracking and order entry system dedicated entirely to its Web catalog; the rest of the companies could generally not distinguish between an order generated by the catalog and an order generated through other marketing means.

3.7 Factors which Influence the Success of Catalog

Of the factors identified in the current survey, two stood out as having the largest impact on the number of monthly visits a catalog is likely to receive: the gross dollars invested in the catalog, both initially and on an ongoing basis, and the placement of the catalog on the World Wide Web, that is, whether the catalog is standalone, inside a mall or embedded in a corporate Web site.

Figure 7 shows a linear relationship between the level of overall investment and the number of visits, that is, the more money invested in the catalog, the more visits the catalog was likely to attract. Further analysis, however, does not

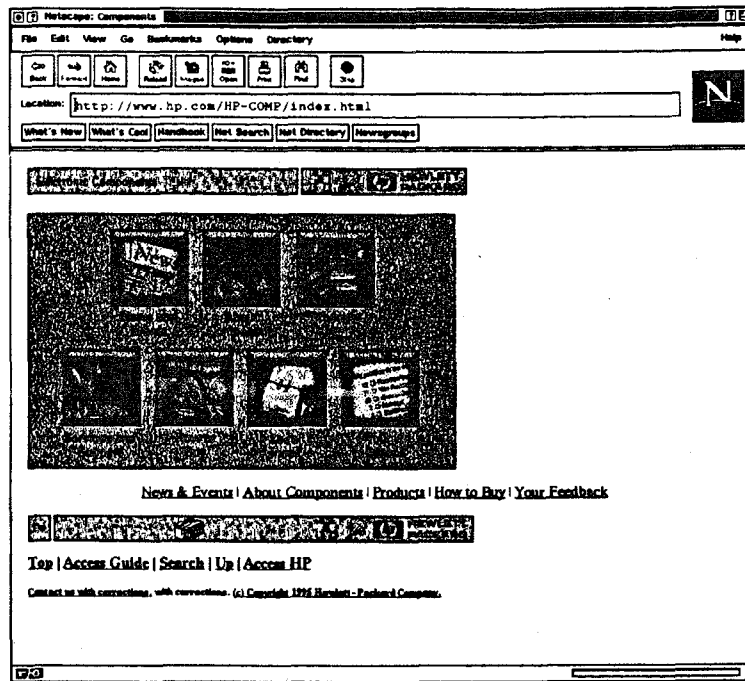


Figure 3: An Example of the Embedded Catalog: Hewlett-Packard.

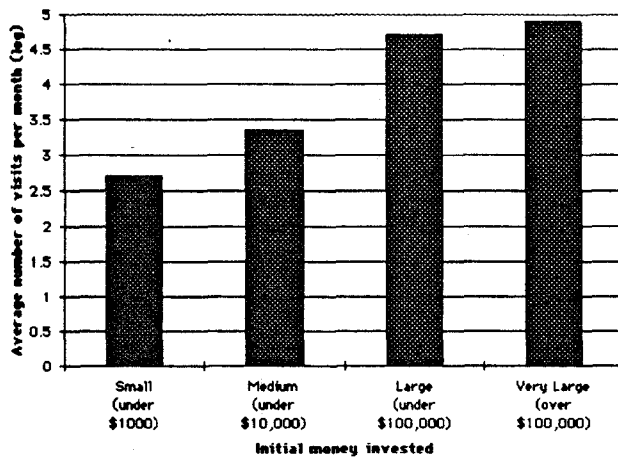


Figure 7: Number of Visits Per Month as a Function of the Dollars Invested

reveal that a similar relationship exists between the investment in any individual cost component (i.e., hardware, software, ongoing telecommunications links, and content creation) and the number of visits. But nearly all of the companies which spent in excess of \$100,000 on their Web sites were found to have a dedicated, full-time Webmaster, and sometimes even a dedicated catalog team, that also received additional cross-functional support from departments such as marketing, sales, and customer support. This might imply that having full-time staff is essential to the success of catalogs.

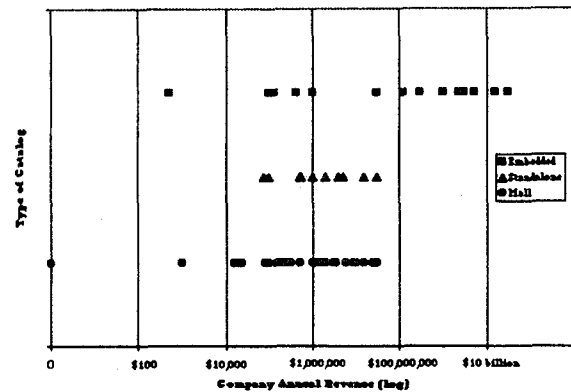


Figure 8: Effect of the Catalog Type on the Number of Visits Per Month.

Figure 8 depicts the relationship between the type of catalogs (i.e., standalone, mall, or embedded) and the revenue size of a company. The data shows that large companies and small Internet services providers (i.e., the small cluster on the top left) are more likely to have embedded catalogs, while small and medium-sized companies tend to have mall or standalone catalogs. The pattern illustrates the importance of being able to attract users to a catalog site and to retain those users over time. Because the Web is becoming so vast, small, little-known companies are likely to remain little-known or "lost" in the cyberspace unless they choose to become part of larger entities, such as electronic malls, which may more easily establish their brand identity in the Web world. However, there is also another group of small com-

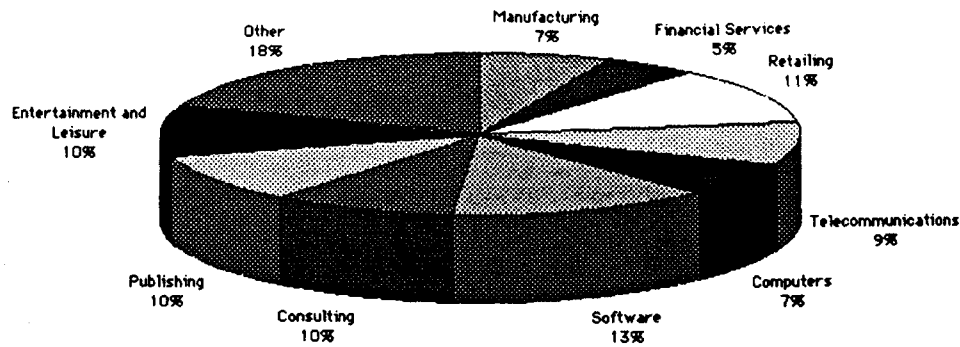


Figure 5: Distribution of Electronic Catalogs by Industry Type.

panies, such as Virtual Vineyards described in Section 2.3, which choose to be standalone. These catalogs distinguish themselves from the crowd through their specialty contents and unique and often sophisticated services.

3.8 Toward a New Measure of Catalog Success: Price Performance Index

In this section, we introduce a new measure of catalog success: the number of visits per dollar invested, or price performance index, which represents the ratio between the number of visits per month and the total investment level in the catalog. The advantage of this measure is that it represents a more accurate indicator of cost-effectiveness, and that it provides a benchmark of price performance which allows comparison with other forms of advertising, as shown in Figure 9. According to [Car95], Netscape is selling a guaranteed 1.5 million "impressions" for a flat \$30,000 fee, which is a price performance of 50 visitors per dollar spent on advertising. While the Netscape advertising is not quite the same as visit to a catalog site, it does provide a benchmark of other forms of electronic publicity. The comparison indicates that, while it is possible to achieve a relatively good price performance on a catalog, electronic catalogs might not be as cost-effective an advertising tool as other venues, such as Netscape.

Visitors per Dollar	Percentage of Catalogs
10-30	12.5%
5-10	12.5%
1-5	22.5%
Under 1	50%
50	Netscape

Figure 9: Distribution of Survey Respondents in Terms of Price Performance Index Benchmarked against Netscape.

4 Conclusions and Future Research Issues

4.1 Conclusions

Electronic catalogs are the front-end to the rapidly evolving Internet-based global marketplace. They are also the

electronic gateway to a company through which customers can obtain product information, order goods and services, make payment, access customer support, and provide feedback around the clock and from anywhere in the world. As a versatile tool, companies can use electronic catalogs to augment existing advertising, marketing, sales, distribution, and support channels. Like other electronic commerce applications, electronic catalogs are still an emerging phenomenon, and there are many basic, unanswered questions, such as what are the main characteristics of the technology? How is the technology currently being used?

This paper was intended to answer these and other related questions on electronic catalogs. It does so by providing a technology overview, and presenting the results of our three-phase survey study of over 100 companies. The main findings from that study are highlighted below:

- Companies are most successful in using electronic catalogs as a means of learning about the Internet technology and enhancing corporate image, but are less successful in selling products or services. However, it seems that selling rates similar to those traditional catalogs can be achieved with relatively low cost.
- The cost of establishing an electronic catalog with basic functionality is relatively low, which might explain why many companies do not think it necessary to justify their investments.
- Functionally, companies place more emphasis on professional appearance and two-way communication capabilities of electronic catalogs but less emphasis on multimedia content, payment security, and advanced search capabilities.
- Marketing and sales departments are main users of electronic catalogs in the company, but top management is among the primary initiators.
- While the two most important measures of catalog success are the number of visits per month and the percentage of visits resulting in sales, it is the number of monthly visits that is most widely used.
- There are two factors that seem to have noticeable impact on the number of monthly visits: the level of overall investment in the catalog and the type of catalogs, i.e., standalone, mall, or embedded.

It should be noted, however, that electronic catalogs are a fast-moving target. The study reported in this paper provides only a snapshot that is limited by its descriptive nature and small sample size. Its results therefore should be interpreted with caution.

4.2 Future Research Issues

Since electronic catalogs are still in their infancy, many important issues remain to be addressed. At the top level, those issues can be grouped into two categories: *technology* and *business practice*.

At the technology front, existing catalogs are still quite primitive in functionality. For example, most catalogs contain static HTML text and do not support dynamic searching. They are standalone rather than integrated with the related product databases. Similarly, catalog functions such as online ordering and payment, if they are available at all, are not integrated with corresponding legacy systems (e.g., ordering, accounts receivable). This lack of back-end represents a serious barrier to the formation of a seamless electronic marketplace.

One other technological issue concerns advanced searching capabilities, such as cross-searching multiple heterogeneous databases, customized searches based on product specific criteria. Much of CommerceNet's research efforts on smart and virtual catalogs are devoted to these issues [Kel95, WK94].

In terms of business practice, one key issue is to identify and develop objective and trackable measures that are appropriate for a particular business objective. Existing measures, such as the number of visits and the percentage of visits resulting in sales, are inaccurate. Moreover, they are not applicable to all business objectives, such as enhancing corporate image. This search for better measures may ultimately require the enhancement of electronic catalog technologies so that the usage patterns of online visitors can be tracked at a fine-grain level.

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References

- [Car95] Jeremy Carl. Rushing to Fill Gap Left by Hits. *Web Week*, August 1995.
- [Cro94] Mary Cronin. *Doing Busines on the Internet: how the electronic highway is transforming American companies*. Van Nostrand Reinhold, 1994.
- [EE95] Jill Ellsworth and Matthew Ellsworth. *Marketing on the Internet: Multimedia Strategies for the World Wide Web*. John Wiley & Sons, 1995.
- [Gup95] Sunil Gupta. Consumer survey of WWW users: preliminary results from the 3rd survey. June 1995.
- [HN94] Donna Hoffman and Thomas Novak. Marking in hypermedia computer-mediated environments: conceptual foundations. Technical report, Owen Graduate School of Management, Vanderbilt University, December 1994.
- [Kel95] Author Keller. Smart catalogs and virtual catalogs. Computer Science Department, Stanford University, 1995.
- [KGSS94] Arthur Keller, Michael Genesereth, N. Singh, and M. Syed. A smart catalog and brokering architecture for electronic commerce. 1994.
- [Mat95] Michael Mathiesen. *Marketing on the Internet*. Maximum Press, Gulf Breeze, FL, 1995.
- [MYB87] Thomas Malone, Joanne Yates, and Robert Benjamin. Electronic markets and electronic hierarchies. *Communications of the ACM*, 30(6):484-96, June 1987.
- [NNP⁺] Robert Neches, Anna-Lena Neches, Paul Postel, Jay M. Tenenbaum, and Robert Frank. Electronic commerce on the Internet.
- [Sok95] Phyllis K. Sokol. *From EDI to Electronic Commerce: a Business Initiative*. McGraw-Hill, Inc., 1995.
- [SWB⁺95] Arie Segev, Dadong Wan, Carrie Beam, Burt Toma, and David Weirod. Financial EDI over the Internet: a case study. CITM Working Paper CITM-95-WP-1006, Fisher Center for Information Technology and Management, University of California, Berkeley, August 1995.
- [TMSW95] Jay Tenenbaum, Cathy Medich, Allan Schiffman, and William Wong. CommerceNet: spontaneous electronic commerce on the Internet. In *COMPCON'95: technologies for the information superhighway*, pages 38-43. IEEE Computer Society Press, March 1995.
- [WK94] William Wong and Arthur Keller. Developing an Internet presence with online electronic catalogs. Workshop on Electronic Commerce, December 1994.